

Data Mining Unit 1 – 69 Unique MCQs (Repeated Removed)

Prepared with all options and ✓ mark for the correct answer.

Note: Q9 (interval vs ratio scale) is included exactly as given, but the options are conceptually ambiguous in theory.

1. Which type of data is commonly mined in data mining?

- Structured data
- Unstructured data
- Semi-structured data
- ✓ All of the above

2. Which of the following is a type of pattern that can be discovered through data mining?

- Associations
- Classification
- Clusters
- ✓ All of the above

3. Which of the following best explains the advantage of data mining in predictive analytics?

- It simplifies data entry processes.
- ✓ It provides insights for future trends and behaviors.
- It eliminates the need for data validation.
- It focuses solely on historical data analysis.

4. Which of the following processes is not typically part of data mining?

- Data cleaning
- Data integration
- Data Transformation
- ✓ Data Entry

5. Which of the following is a significant challenge when mining web data?

- The structured nature of web data
- The small volume of web data
- ✓ The heterogeneity and high dimensionality of web data
- The lack of user interaction data

6. Which of the following best exemplifies an attribute that can be both discrete and ordinal?

- Age in years
- Temperature in Celsius
- ✓ Customer satisfaction rating (e.g., very dissatisfied, dissatisfied, neutral, satisfied, very satisfied)
- Height in centimeters

7. Which of the following is the most effective measure of the center of symmetric data set?

- Mode
- Midrange
- ✓ Mean

- Median

8. In data mining, why is it essential to understand the business domain before analyzing the data?

- To ensure the use of the latest algorithms
- ✓ To translate business objectives into specific data mining goals
- To optimize the hardware for data processing
- To simplify the data cleaning process

9. Which of the following is true for interval-scaled attributes but not for ratio-scaled attributes?

- They have a meaningful zero point.
- The difference between values is meaningful.
- The ratio between values is meaningful.
- The order of values is meaningful.

10. If a survey includes questions where respondents rate their satisfaction on a scale of 1 to 5, what type of attribute is the rating?

- Nominal
- ✓ Ordinal
- Interval
- Ratio

11. Which statistical measure is most appropriate for summarizing ordinal data?

- Mean
- ✓ Median
- Standard deviation
- Range

12. Which measure describes the extent to which a distribution is asymmetric about its mean?

- ✓ Skewness
- Kurtosis
- Variance
- Standard deviation

13. Which of the following is an example of a ratio-scaled attribute?

- Temperature in Celsius
- Customer satisfaction rating
- ✓ Weight in kilograms
- Level of education

14. Which of the following refers to the set of features that describe a data object?

- ✓ Attribute vector
- Instance
- Sample
- Data point

15. What is the main objective of regression analysis in data mining?

- To classify data into different categories
- ✓ To find relationships between variables and predict continuous outcomes
- To reduce the number of dimensions in a dataset
- To cluster similar records together

16. Which measure of variability is calculated by taking the difference between the third quartile (Q3) and the first quartile (Q1)?

- Variance
- Standard deviation
- ✓ Interquartile range
- Range

17. Which measure of central tendency is calculated by arranging the data in ascending order and selecting the value at the midpoint?

- Mean
- ✓ Median
- Mode
- Range

18. Which distance measure is calculated as the sum of the absolute differences between the coordinates of two points?

- Euclidean distance
- ✓ Manhattan distance
- Minkowski distance
- Hamming distance

19. A nominal attribute with two categories is known as _____

- Attribute vector
- Numeric attribute
- ✓ Binary attribute
- Ordinal attribute

20. The attribute type with values that are measurable quantities represented by integers or real values is called as _____

- Nominal
- Binary
- ✓ Numeric
- Ordinal

21. Which of the following is true about the range of a set of values of a data set?

- It is the product of the largest and the smallest value
- It is the average of the largest and the smallest value
- It is the sum of the largest and the smallest value
- ✓ It is the difference between the largest and the smallest value

22. The lines outside the box, extending to the highest and lowest values in a box plot are called as _____

- Medians
- Quartiles
- Quintiles
- ✓ Whiskers

23. Which of the following is a step in the KDD process?

- ✓ Data cleaning
- Data warehousing
- Data aggregation
- Data scaling

24. Which of the following is an example of categorical data?

- Height
- Weight
- Age
- ✓ Gender

25. Subjective measures of interestingness depend on:

- Probability theory
- Support and confidence
- ✓ User's background knowledge and expectations
- Entropy calculations

26. In the context of data mining, what does the term 'noise' refer to?

- Meaningful patterns in the data
- ✓ Unnecessary or irrelevant data
- Large volumes of data
- Data stored in a structured format

27. Which of the following describes the term 'pattern evaluation' in data mining

- The process of extracting patterns from data
- ✓ The process of identifying the interestingness of patterns
- The process of storing data
- The process of deleting irrelevant patterns

28. What is the role of a data warehouse in data mining?

- ✓ To provide a platform for data storage and management
- To analyze and interpret data
- To visualize data
- To delete unnecessary data

29. Which of the following is a shared concern between (Business Intelligence)BI and Web Mining applications?

- Optimizing hardware

- Managing backup files
- ✓ Extracting useful patterns from large, dynamic data sources
- Designing user interfaces

30. Which of the following fields contributes techniques like regression, correlation, and hypothesis testing to data mining?

- Artificial Intelligence
- ✓ Statistics
- Visualization
- Operating systems

31. What does the term 'data discretization' mean in data mining?

- ✓ Transforming continuous data into categorical data
- Combining multiple data sources into one
- Scaling data to a specific range
- Removing noise from data

32. Pick the wrong data mining functionality among the given data mining functionalities.

- Classification
- Clustering
- Regression
- ✓ Object Description

33. Which of the following is not true about scatter plots?

- ✓ It is used in the case of univariate distribution
- It is used to identify relationship between attributes
- It is used to identify clusters
- It is used to identify outliers

34. What is a significant ethical concern related to data mining?

- High computational cost
- Data redundancy
- ✓ Privacy invasion
- Data normalization

35. Which attribute type would be used to represent a unique identifier such as a Social Security Number?

- ✓ Nominal
- Ordinal
- Interval
- Ratio

36. Which measure is the square root of the variance?

- Mean
- Median
- ✓ Standard deviation

- Interquartile range

37. What is the measure of central tendency that is most sensitive to extreme values in a dataset?

- ✓ Mean
- Median
- Mode
- Interquartile range

38. What is the term for the measure that divides a data set into four equal parts?

- Mean
- Median
- ✓ Quartile
- Percentile

39. For a data object stored in a database, the attributes of the data object are represented by _____

- ✓ Columns of the database
- Rows of the database
- Relation schema
- Relational index

40. Which of the following does not describe the properties of a data object?

- Attributes
- Features
- Dimensions
- ✓ Instances

41. The binary attributes with both the categories having equal importance are called as _____

- ✓ Symmetric
- Asymmetric
- Asymmetric
- Numeric

42. Which of the following cannot be appropriately represented as a continuous attribute?

- Height
- Weight
- Time
- ✓ PIN codes

43. The measure of central tendency that separates the data set into higher and lower halves is _____

- Mean
- ✓ Median
- Mode
- Midrange

44. Which of the following is not true about mode?

- It is the most frequently occurring value
- It is defined for both qualitative and quantitative values
- ✓ A data set can have at most one mode
- A data set can have more than one mode

45. Which of the following is the most appropriate sequence for the given negatively skewed data

- Mean, mode, median
- ✓ Mode, median, mean
- Mean, median, mode
- Mode, mean, median

46. Which of the following are the components of five-number summary in the case of skewed data?

- ✓ Minimum, Q1, Median, Q3, Maximum
- Minimum, Q1, Mode, Q3, Maximum
- Minimum, Q1, Mode, Q3, Maximum
- Minimum, Q1, Mode, Q2, Maximum

47. Which of the following is true if the value of a dissimilarity measure is zero for two objects?

- ✓ The two objects are very similar
- The two objects are very dissimilar
- The two objects are moderately dissimilar
- The two objects are moderately similar

48. Which data mining technique involves grouping similar records into clusters?

- Classification
- ✓ Clustering
- Regression
- Association rule mining

49. Which visualization technique is often used in cluster analysis to show groupings in data?

- ✓ Scatter plot with color-coded clusters
- Pie chart
- Line graph
- Box plot

50. Summarization is also known as _____

- ✓ Characterization
- Link analysis
- Affinity analysis
- Series analysis

51. What is a dissimilarity matrix used for in data mining?

- To store data
- To measure the similarity between data points

- To visualize data
- ✓ To measure the distance or dissimilarity between data points

52. _____ predicts future trends & behaviors, allowing business managers to make proactive, knowledge-driven decisions.

- Data warehouse.
- ✓ Data mining.
- Datamarts.
- Metadata.

53. Which measure of central tendency is least affected by outliers?

- Mean
- ✓ Median
- Mode
- Range

54. Which of the following metrics is commonly used to measure the dissimilarity between two numerical data points?

- ✓ Euclidean distance
- Mode
- Median
- Kurtosis

55. Which of the following best describes 'knowledge discovery in databases' (KDD)?

- ✓ The overall process of converting raw data into useful information.
- The process of storing large volumes of data.
- The process of deleting redundant data.
- The process of predicting future data.

56. The issues of “Scalability and efficiency of the data mining algorithms” come under:

- User Interaction and Mining Methodology Issues
- Diverse Data Types Issues
- ✓ Performance Issues
- None of the above

57. Objective measures of pattern interestingness are based on:

- User beliefs
- Surprise factor
- ✓ Statistical properties of the data
- Business requirements

58. Which task is most commonly associated with data mining in a web search engine?

- Keyword coloring
- Page rendering
- ✓ Ranking and relevance analysis

- PDF export

59. After mining useful patterns from data, which step helps to clearly show or explain the results?

- Data Cleaning
- Data Integration
- ✓ Pattern visualization
- Network packet routing

60. What is a major issue in data mining?

- Data privacy
- Data quality
- Scalability
- ✓ All of the above

61. Which data mining technique is primarily used for predicting a categorical label?

- Regression
- Data scaling
- ✓ Classification
- Association rule mining

62. What happens during the data selection step in KDD?

- Irrelevant attributes are removed
- The final results are interpreted
- ✓ A target dataset is identified from the entire database
- Patterns are visualized for business users

63. Machine learning contributes to data mining by providing

- Physical storage systems
- ✓ Pattern discovery and prediction algorithms
- Software licensing models
- Internet protocols

64. The ability of data mining systems to visually present data patterns is supported by:

- Image processing
- ✓ Data visualization technologies
- Optical character recognition
- Cybersecurity tools

65. Which of the following visualization types is most effective for distribution analysis of numerical attributes?

- ✓ Histogram
- Scatter plot
- Pie chart
- Tree map

66. 100 quantile means?

- ✓ Percentiles
- 100 highest values in data
- Value appears 100 times
- Data with 100 rows

67. In data mining, which of the following is an attribute type?

- Nominal
- Ordinal
- Interval
- ✓ All of the above

68. Which of the following is true about the arithmetic mean?

- ✓ It is sensitive to outliers
- It is not sensitive to outliers
- Extreme values do not affect the mean
- Mean of the data cannot be corrupted

69. In the KDD process, which step is responsible for removing noise and inconsistent data?

- Data mining
- Data transformation
- ✓ Data cleaning
- Evaluation